

Muhammad Shaheer Luqman

Driven and dedicated Software Engineer, experienced in building and optimizing production-grade systems. Strong in problem-solving, system design, and team collaboration.

Work Experience

Applied Research Engineer – Retrocausal (Intern → Full-Time)

Mar 2024 - Present

- Developed an end-to-end **video activity recognition pipeline** using **Body Pose Estimation** and **Vision-Language models** to translate visual and textual inputs into PMTS codes for **industrial process optimization**.
- Designed and implemented **front-end, back-end, and AI components** for **BMW**, collaborating closely from **requirements gathering to final delivery**.
- Worked on **research initiatives** including **unsupervised action segmentation** and **temporal long-video understanding**, contributing experimental insights toward scalable industrial video analytics.
- Applied strong **software engineering practices** to build modular, maintainable systems, including **LLM-based prompt engineering** and **automated Excel report generation** with visual analytics, leading to ownership of **16 modules** due to consistent high-quality deliverables.

Educational Background

Bachelor of Science in Computer Science (BS-CS)

NUCES FAST | Sep 2021 - Aug 2025 | CGPA: 3.90/4.00

A Levels

Generation's School | Sep 2019 - Aug 2021 | Grades: A*, 2A

Publications

Ensuring Child Safety: An IoT-Based Surveillance System for Remote Monitoring and Detection of Anomalous Behavior

2024 International Conference on Emerging Trends in Smart Technologies (ICETST), Karachi, Pakistan

Publisher: IEEE | Journal: IEEE Xplore

<https://ieeexplore.ieee.org/document/10737980>

Projects

Digital Marketing Tool Using GenAI - PixelForge | Sep 24 - May 25

React | Flask | Stable Diffusion | Video/Image Ads Generation

Multiple Websites/WebApps | Sep 24 - Apr 25

MERN | Project Lead | Full Stack | Automation Scripts

Crypto Trading Simulator WebApp (MERN) | Sep 23 - Nov 23

MERN | API integration | AntDesign Frontend | Database

Competitive Programming | Feb 22 – Feb 25

Problem-solving | Algorithms | Data Structures

Honors and Awards

- **Gold Medal** - ICPC National Competitive Programming - Nov 24
- **Bronze Medal** - ICPC National Competitive Programming - Feb 24
- **Best Team Award** - Coder's Cup, ACM NUCES, FAST - Oct 23
- **4 Rector's List Award** - Full 4.0 GPA in 1st, 4th, 7th, 8th Semester
- **Winner** - Code Sprint Competition - PROCOM 23 - Mar 23
- **4 Dean's List Award** – 3.5+ GPA in 2nd, 3rd, 5th, 6th Semester

Contact

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github.com/ShaherLuqman

Skills

Technical Skills

- Artificial Intelligence (Scikit Learn, Huggingface, PyTorch)
- Computer Vision (Mediapipe, Yolo, Diffusion Models)
- NLP (Whisper, BERT, 11Labs)
- LLM Prompt Engineering
- Databases (MySQL/Mongo/Oracle)
- Backend (Django, Express. Js)
- Web Frontend (React, Next, Vue)
- Data Analysis (NumPy, Matplotlib)
- Web Automation (Selenium)
- Microsoft Assembly Language
- Arduino Programming
- Mobile Development (React Native)

Frameworks

- Django
- Flask
- FastAPI
- C/C++
- React/Next
- React Native
- Express

Certification/Short Courses

- Supervised Mach. Learning - DeepLearning.AI
- AI For Everyone - DeepLearning.AI
- Introduction to Generative AI - Google

Volunteer Experience

- **Director Technical Affairs**, Developers Day'25 | Jan 25 - Jun 25
- **Technical Lead**, ACM NUCES | Aug 24 - Jun 25
- **CS Competition Head**, Developers Day'24 | Jan 24 – May 24
- **CS Competition Cohead**, Procom'24 | Oct 23 – May 24
- **Training and Mentorship Head**, ACM NUCES | Sep 23 – Jan 24